

Two Positive Subcases Of The Complex Real-Form Question For Complemented Subspaces Of Complex Banach Lattices

Run fa_banach_001, agent agent_lane_06

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Abstract

De Hevia, Martinez-Cervantes, Salguero-Alarcon, and Tradacete ask in Remark 5.5 of *A counterexample to the complemented subspace problem in Banach lattices* whether every complemented subspace of a complex Banach lattice is linearly isomorphic to the complexification of some real Banach space. We record several positive subcases and normalization criteria. First, every finite-codimensional complex subspace of a complex Banach lattice has such a real form. Second, any complex Banach space with an unconditional Schauder basis has such a real form. Third, an anti-linear automorphism can be normalized into a conjugation whenever the square has a suitable holomorphic square root. This remains true after splitting off finitely many bad spectral values with finite-dimensional Riesz subspaces, and hence in an important compact-perturbation case. In particular, the same conclusion holds for the range of any projection which is sufficiently close to its conjugate projection. The full infinite-codimensional question remains open.

1 Source Question

The source paper solves the complemented subspace problem for real and complex Banach lattices in the negative. At the end of Section 5, the authors isolate a remaining question:

Regarding the CSP for complex Banach lattices, one question that remains open is whether every complemented subspace of a complex Banach lattice is linearly isomorphic to the complexification of some real Banach space.

They also point out that a complexification is necessarily isomorphic to its complex conjugate, while the known examples not isomorphic to their conjugates fail Gordon–Lewis local unconditional structure and hence cannot be complemented subspaces of complex Banach lattices.

2 Definitions

A *conjugation* on a complex Banach space Z is a bounded anti-linear operator $J : Z \rightarrow Z$ such that $J^2 = I$. The canonical example is the conjugation on the complexification $E_{\mathbb{C}}$ of a real Banach space E .

We say that a complex Banach space Y has a *real form* if it is complex-linearly isomorphic to the complexification of a real Banach space.

3 Proof Intuition

The source question asks whether a complemented complex subspace must remember some real structure. A real structure is exactly a conjugation up to isomorphism. Thus the strategy is to manufacture a bounded anti-linear involution on the subspace.

For finite codimension, this is possible because the defining quotient map has finite-dimensional range. One can choose a finite-dimensional complement spanned by vectors from the ambient real part, so the complement is invariant under conjugation. Projecting the ambient conjugation back along this complement gives a conjugation on the subspace.

For a space with an unconditional basis, the conjugation is even more concrete: write a vector in coordinates and conjugate the coefficients. The unconditionality constant controls all unimodular coordinate multipliers, so this coefficient conjugation is bounded.

The next obstruction is spectral rather than geometric. If a space is isomorphic to its complex conjugate, it carries an anti-linear automorphism A . This is not automatically a conjugation, since A^2 may be a nontrivial complex-linear automorphism. When A^2 has a square root chosen compatibly with conjugation, functional calculus renormalizes A to an anti-linear involution.

For arbitrary codimension, a projection P onto the subspace need not commute with the ambient conjugation J . However, if P is close to its conjugate JPJ , the anti-linear map PJ on the range has square close to the identity. Taking a bounded square root of this square normalizes PJ to an actual involution.

4 A Basic Real-Form Lemma

Lemma 1. *Let Y be a complex Banach space. If Y admits a bounded conjugation C , then Y is complex-linearly isomorphic to the complexification of a real Banach space.*

Proof. Let $E = \text{Fix}(C) = \{y \in Y : Cy = y\}$, regarded as a real Banach space with the inherited norm. Equip $E_{\mathbb{C}}$ with the usual complexification norm

$$\|e + if\|_{E_{\mathbb{C}}} = \sup_{\theta \in \mathbb{R}} \|e \cos \theta + f \sin \theta\|_E.$$

Every $y \in Y$ has the unique decomposition

$$y = \frac{y + Cy}{2} + i \frac{y - Cy}{2i},$$

with both summands inside E . Thus the natural map

$$E_{\mathbb{C}} \ni e + if \mapsto e + if \in Y$$

is a complex-linear bijection. It is bounded since $\|e + if\|_Y \leq \|e\| + \|f\| \leq 2\|e + if\|_{E_{\mathbb{C}}}$. Its inverse is bounded because the two real components above are controlled by $(1 + \|C\|)\|y\|/2$, while $\|e + if\|_{E_{\mathbb{C}}} \leq \|e\| + \|f\|$. Therefore it is an isomorphism. $\square$

5 Finite-Codimensional Subspaces

Theorem 1. *Let E be a real Banach space and let $Z = E_{\mathbb{C}}$ be its complexification, with canonical conjugation J . If $Y \subset Z$ is a closed complex subspace of finite codimension, then Y is complex-linearly isomorphic to the complexification of a real Banach space.*

Proof. Let $m = \dim_{\mathbb{C}} Z/Y$. The case $m = 0$ is immediate, so assume $m > 0$. Choose a bounded surjective complex-linear map

$$F : Z \rightarrow \mathbb{C}^m$$

with kernel Y . The complex span of $F(E)$ is all of $\mathbb{C}^m$, because every element of $Z = E + iE$ is a complex linear combination of elements of E . Hence we may choose $u_1, \dots, u_m \in E$ such that $F(u_1), \dots, F(u_m)$ is a complex basis of $\mathbb{C}^m$.

Put $M = \text{span}_{\mathbb{C}}\{u_1, \dots, u_m\}$. Then $M \cap Y = \{0\}$ and $Z = Y \oplus M$. Since each u_j belongs to the real part E , the subspace M is invariant under J . Let $R : Z \rightarrow Y$ be the bounded projection associated with the direct sum $Z = Y \oplus M$.

Define $C : Y \rightarrow Y$ by

$$Cy = R(Jy).$$

This map is bounded and anti-linear. We claim that $C^2 = I_Y$. Indeed, for $y \in Y$, write $Jy = Cy + m$ with $m \in M$. Applying J gives

$$y = J(Cy) + Jm.$$

Since M is J -invariant, $Jm \in M$. Applying the projection R to the last equality yields

$$y = R(J(Cy)) = C(Cy).$$

Thus C is a bounded conjugation on Y . Lemma 1 completes the proof. $\square$

Corollary 1. *Every finite-codimensional complex subspace of a complex Banach lattice is linearly isomorphic to the complexification of a real Banach space.*

Proof. By definition, a complex Banach lattice is the complexification of a real Banach lattice. Apply Theorem 1. $\square$

6 The Unconditional-Basis Subcase

Theorem 2. *Let Y be a complex Banach space with an unconditional Schauder basis. Then Y is complex-linearly isomorphic to the complexification of a real Banach space.*

Proof. Let (e_n) be an unconditional basis for Y . There is a constant K such that every coordinate multiplier

$$M_\lambda \left(\sum_n a_n e_n \right) = \sum_n \lambda_n a_n e_n, \quad |\lambda_n| \leq 1,$$

defines a bounded complex-linear operator with norm at most K .

Define

$$C \left(\sum_n a_n e_n \right) = \sum_n \overline{a_n} e_n.$$

For a fixed vector $y = \sum_n a_n e_n$, choose $\lambda_n = \overline{a_n}/a_n$ when $a_n \neq 0$, and $\lambda_n = 0$ otherwise. Then $Cy = M_\lambda y$, so $\|Cy\| \leq K\|y\|$. Thus C is bounded. It is anti-linear and satisfies $C^2 = I$. Lemma 1 now gives the conclusion. $\square$

Corollary 2. *Any complemented subspace of a complex Banach lattice which has an unconditional Schauder basis is linearly isomorphic to the complexification of a real Banach space.*

Remark 1. This includes the basis cases but not arbitrary unconditional finite-dimensional decompositions. Kalton's spaces $Z_2(\alpha)$, which are not isomorphic to their complex conjugates, show why block decompositions alone are a delicate frontier rather than an automatic real-form criterion.

7 A Spectral Normalization Criterion

Theorem 3. *Let Y be a complex Banach space. Suppose that $A : Y \rightarrow Y$ is a bounded invertible anti-linear operator. If the spectrum of the complex-linear operator $L = A^2$ is contained in $\mathbb{C} \setminus (-\infty, 0]$, then Y is complex-linearly isomorphic to the complexification of a real Banach space.*

Proof. Since A is anti-linear and $L = A^2$, we have $AL = LA$. The spectral assumption lets us define the principal inverse square root

$$S = L^{-1/2}$$

by holomorphic functional calculus on the slit plane $\mathbb{C} \setminus (-\infty, 0]$. The principal branch satisfies

$$f(\bar{z}) = \overline{f(z)}$$

for $f(z) = z^{-1/2}$. By approximating f uniformly on a conjugation symmetric neighborhood of $\sigma(L)$ by polynomials with real coefficients, or equivalently by the symmetry of the Cauchy integral formula, one obtains

$$AS = SA.$$

Now put $C = SA$. Then C is bounded and anti-linear, and

$$C^2 = SASA = S^2 A^2 = L^{-1} L = I.$$

Thus C is a bounded conjugation. Lemma 1 completes the proof. $\square$

Corollary 3. *Let Z be a complex Banach space with a bounded conjugation J , let $P : Z \rightarrow Z$ be a bounded complex-linear projection, and set $Y = PZ$. If*

$$A = PJ|_Y : Y \rightarrow Y$$

is invertible and $\sigma(A^2) \subset \mathbb{C} \setminus (-\infty, 0]$, then Y is linearly isomorphic to the complexification of a real Banach space.

Proof. Apply Theorem 3. $\square$

Theorem 4. *Let Y be a complex Banach space and let $A : Y \rightarrow Y$ be a bounded invertible anti-linear operator. Put $L = A^2$. Suppose that*

$$B = \sigma(L) \cap (-\infty, 0]$$

is finite and that every point of B is an isolated spectral value whose Riesz spectral subspace is finite-dimensional. Then Y is complex-linearly isomorphic to the complexification of a real Banach space.

Proof. If $B = \emptyset$, this is exactly Theorem 3. Assume $B \neq \emptyset$.

Let P_B be the Riesz spectral projection of L associated with B . Set $E = P_B Y$ and $F = (I - P_B)Y$. By hypothesis E is finite-dimensional, and

$$Y = E \oplus F$$

as a topological direct sum of complex subspaces, and the spectrum of $L|_F$ is contained in $\mathbb{C} \setminus (-\infty, 0]$.

The spaces E and F are invariant under A . Indeed, for every $z \in \mathbb{C}$,

$$A(zI - L) = (\bar{z}I - L)A.$$

Choose the contours defining P_B symmetric with respect to complex conjugation. The Riesz integral and the anti-linearity of A then give $AP_B = P_B A$. Therefore $A|_F$ is a bounded anti-linear automorphism of F , and Theorem 3 gives a bounded conjugation on F .

The finite-dimensional space E also has a bounded conjugation, by choosing any complex basis and conjugating coefficients. Taking the direct sum of these two conjugations gives a bounded conjugation on $Y = E \oplus F$. Lemma 1 completes the proof. $\square$

Corollary 4. *Let Y be a complex Banach space and let $A : Y \rightarrow Y$ be a bounded invertible anti-linear operator. Suppose*

$$A^2 = \lambda I + K,$$

where K is compact and $\lambda \in \mathbb{C} \setminus (-\infty, 0]$. Then Y is complex-linearly isomorphic to the complexification of a real Banach space.

Proof. Let $L = A^2$. The spectrum of the compact perturbation $L = \lambda I + K$ is the point λ , together with at most countably many isolated eigenvalues of finite algebraic multiplicity which can accumulate only at λ . Since $\lambda \notin (-\infty, 0]$, the bad spectral set $\sigma(L) \cap (-\infty, 0]$ is finite, and all its Riesz spectral subspaces are finite-dimensional. Apply Theorem 4. $\square$

Corollary 5. *Let Y be a complex Banach space in which every bounded complex-linear operator is of the form $\mu I + K$, with K compact. If Y is anti-linearly isomorphic to itself and some anti-linear automorphism A satisfies*

$$A^2 = \lambda I + K, \quad \lambda \notin (-\infty, 0],$$

then Y has a real form.

Consequently, any non-real-form space of scalar-plus-compact type which is anti-linearly isomorphic to itself must have only the remaining “quaternionic” obstruction: every such anti-linear square has scalar part on the non-positive real axis.

Proof. The first assertion is Corollary 4. The final sentence is just the contrapositive applied to anti-linear automorphisms of Y . $\square$

8 A Perturbative Projection Criterion

Theorem 5. *Let Z be a complex Banach space with a bounded conjugation J . Let $P : Z \rightarrow Z$ be a bounded complex-linear projection and set $Y = PZ$. If*

$$\|P\| \|P - JPJ\| < 1,$$

then Y is complex-linearly isomorphic to the complexification of a real Banach space.

Proof. Put $Q = JPJ$. Then Q is a complex-linear projection onto JY . Define the bounded anti-linear map

$$A : Y \rightarrow Y, \quad Ay = PJy.$$

Its square is the bounded linear operator

$$L = A^2 = PQ|_Y.$$

For $y \in Y$, since $Py = y$,

$$\|Ly - y\| = \|P(Q - P)y\| \leq \|P\| \|Q - P\| \|y\| < \|y\|.$$

Thus $\|L - I_Y\| < 1$. In particular L is invertible, and the binomial series gives a bounded inverse square root

$$S = L^{-1/2}$$

inside the closed algebra generated by L . Moreover A is injective because $A^2 = L$ is invertible, and A is surjective since for any $y \in Y$,

$$y = A(AL^{-1}y).$$

Thus A is an anti-linear automorphism. The binomial coefficients are real, so A commutes with S . Therefore

$$C = SA$$

is bounded, anti-linear, and satisfies

$$C^2 = SASA = S^2A^2 = L^{-1}L = I_Y.$$

So C is a bounded conjugation on Y . Lemma 1 again gives the desired real form. $\square$

Corollary 6. *Let $Z = E_{\mathbb{C}}$ be a complex Banach lattice with its canonical conjugation J . If P is a projection on Z whose range is Y and $\|P\| \|P - JPJ\| < 1$, then Y is linearly isomorphic to the complexification of a real Banach space. In particular this holds whenever $P = JPJ$.*

9 Limitations And Novelty Check

The result is only a partial positive answer to Remark 5.5. It does not settle arbitrary infinite-codimensional complemented subspaces of complex Banach lattices, and it gives no counterexample to the full question. Instead, it shows that any counterexample must avoid finite codimension, must not have an unconditional Schauder basis, and must not admit either a sufficiently conjugation-stable projection or an anti-linear self-isomorphism whose square is spectrally normalizable, even up to compact perturbation away from the non-positive real axis.

The run indexes were searched for arXiv:2310.02196, the title of the source paper, and the phrases “complexification”, “complex conjugate”, “GL-lust”, and “complex Banach lattice”. A web search on June 16, 2026 found the source arXiv page, the 2025 survey by de Hevia and Tradacete on complemented subspaces of Banach lattices, Kalton’s $Z_2(\alpha)$ examples, and Gordon–Junge–Meyer–Reisner’s finite-dimensional analysis of the Kalton–Peck twisted sum, but no exact prior run packet or later answer to Remark 5.5. The arguments above are elementary, so novelty confidence should be treated as modest; the intended contribution is the scoped partial answer and the explicit obstruction profile.

References

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